Introduction to Java & OOP Concepts

1. What is Java?

Java is a high-level, class-based, object-oriented programming language that is designed to have as few implementation dependencies as possible. It is a general-purpose programming language intended to let programmers write once, run anywhere (WORA), meaning that compiled Java code can run on all platforms that support Java without the need to recompile.

Java applications are typically compiled to bytecode that can run on any Java virtual machine (JVM) regardless of the underlying computer architecture.

2. JDK, JRE, and JVM

Understanding the differences between JDK, JRE, and JVM is crucial for any Java developer:

- **JVM (Java Virtual Machine):** It is an abstract machine. It is a specification that provides a runtime environment in which Java bytecode can be executed. It can also run those programs which are written in other languages and compiled to Java bytecode.
- **JRE (Java Runtime Environment):** It is a software package that contains what is required to run a Java program. It includes a Java Virtual Machine implementation together with an implementation of the Java Class Library.
- **JDK (Java Development Kit):** It is a software development environment used for developing Java applications and applets. It includes the JRE, an interpreter/loader (Java), a compiler (javac), an archiver (jar), a documentation generator (Javadoc) and other tools needed in Java development.

3. Features of Java

- **Simple:** Java is easy to learn, and its syntax is clean, crisp, and easy to understand.
- **Object-Oriented:** In Java, everything is an Object. Java can be easily extended since it is based on the Object model.
- **Platform Independent:** Unlike many other programming languages including C and C++, when Java is compiled, it is not compiled into platform-specific machine, rather into platform-independent byte code.
- **Secure:** With Java's secure feature, it enables to develop virus-free, tamper-free systems.

4. OOP Principles in Java

- **Encapsulation:** Wrapping data and code together as a single unit.
- **Inheritance:** Mechanism where one class acquires the properties of another.
- **Polymorphism:** Ability of a variable, function or object to take on multiple forms.
- **Abstraction:** Hiding internal details and showing functionality only.

5. First Java Program and Compilation

A basic Java program looks like this:

```
public class HelloWorld {
public static void main(String[] args) {
System.out.println("Hello, World!");
}
}
```

Compilation: javac HelloWorld.java (produces HelloWorld.class)
Execution: java HelloWorld (runs the bytecode on the JVM).